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Review

Education and Farmers' Environmental Performance: A Systematic Literature Review

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Abstract: Climate change and environmental degradation present significant and ongoing challenges to the agricultural sector. Formal education has been shown to equip farmers with the knowledge and skills needed to improve their environmental performance and address these problems. However, there is a notable gap in the literature regarding a comprehensive review of how farmers' education impacts environmental outcomes. This paper aims to bridge that gap by synthesizing the results of studies exploring the role of education in explaining environmental performance. To this end, we conducted a systematic review of the existing literature. We organised the resulting 127 studies into thematic groups to provide corresponding overviews. Subsequently, we conducted a vote-count analysis of the reported direction and statistical significance of effects and associations between education and environmental performance. We observe that reporting at the variable level rather than the overall study level may have led to the importance of education for specific groups of farmers being overlooked. We suggest that the way education variables are constructed can influence the results of studies, leaving the results of vote-count analysis open to interpretation. The findings of the review provide the requisite evidence base to inform the development of effective policy measures that enhance the sustainability of farming.

Keywords: educational attainment; sustainability; environmental performance

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1. Introduction

Climate change continues to be one of the most significant issues confronting global agriculture, with consequential effects on both food production and the environment [1]. Climate-driven extremes such as irregular rainfall patterns, drought, heatwaves, and floods are already exceeding plant and animal tolerances, with consequential effects on agricultural productivity and farmers' livelihoods [2]. As nations across the world pursue a global target of net-zero carbon emissions by 2050, innovation across products, services, and processes within and beyond the value chain is essential [3]. The agricultural sector will be central to this, with a need for primary producers to adjust their farming systems to achieve more sustainable production that ensures not just the protection of the environment but also enhances or at least maintains farm-level productivity [4]. This will require the development of policies aligned with sustainability goals, promoting practices that reduce emissions and conserve resources, and equipping farmers with the necessary knowledge and skills through education and training.

Global food systems contribute over a third of the total anthropogenic greenhouse gas emissions [5]. The industry's capacity to reduce emissions while maintaining global food

security will depend on farmers' abilities to leverage farm-level innovations and to adopt climate-smart, sustainable agricultural practices [5,6]. Research evidence suggests that attitude towards, intention to adopt, and implementation of environmentally sustainable practices are significantly influenced by an individual's level of engagement in formal learning and their level of educational attainment [7]. For example, at the EU level, although around 60% of farm managers with at least a secondary level of education used soil quality information to decide on soil management, less than 30% of farmers with primary education did so [8].

At an individual level, formal education helps farmers develop the necessary knowledge and skills, enabling them to access, use, and apply relevant information. This allows them to address challenges, adapt to a changing environment, and ultimately allocate their resources efficiently [9]. Although the development of farmers' knowledge and skills can be undertaken through a range of training methods, such as practical training, long-term training, on-site training, and actual demonstrations, increasing the level of formal education among farmers is now considered key to promoting improved farm-level performance [10–13]. Indeed, as highlighted by Lindblom et al. [14], the adoption of more sustainable agricultural systems is closely linked to a farmer's ability to interpret and make decisions around information and data. This being the case, improved environmental performance at the farm level will necessitate a paradigm shift in production and management practices.

Despite the pivotal role of formal agricultural training and continuous professional development in enhancing farmers' education, no comprehensive review of the impact of farmers' education on environmental performance has been undertaken. Several factors may contribute to this gap, including the divergence of environmental indicators associated with performance measurement and the varying forms of farmer education across different countries. For instance, in Europe, only 8.5% of the total farming population has received full agricultural training, while another 28.7% has had some form of agricultural training [15].

Thus, this study aims to fill this gap in the literature. While there are literature review papers on factors influencing environmental performance, such as those by Sapbamrer and Thammachai [16] and Xie and Huang [17], most of these papers focus on factors influencing a specific environmental performance indicator, such as the adoption of organic farming. The objective of this study is to undertake a comprehensive and systematic review of the impact of formal education on environmental performance indicators across several dimensions. While a farmer's environmental performance may be influenced by a range of factors (socioeconomic, institutional, economic, and behavioural), this study will highlight the specific role of education. An understanding of the role of education on farmers' environmental performance will provide the requisite evidence base needed to develop appropriate policy measures to improve the sustainability of farming and combat climate change and its effects.

2. Materials and Methods

In undertaking this study, we adopted a systematic review and meta-analysis (PRISMA) approach [18], which advises on good practices for reporting and structuring systematic reviews.

2.1. Search Strategies

The search strategy initially involved using Web of Science, a peer-reviewed database. Accordingly, we searched for relevant studies using a combination of keywords, while applying the first and second eligibility criteria, as described in the next paragraph, which restricted our interest to peer-reviewed studies published in the 21st century. The

three keywords that were used in each search included (1) "farm"; (2) a variation of terms concerning education: "educational attainment", "agricultural education", "educational level", "education level", "formal education"; and (3) one of the following: "nutrient", "soil", "water", "climate", "conservation", "carbon", "ecosystem services", "environmental efficiency", "eco-efficiency", "environmental performance". Using standalone keywords for "education" and "environment" yielded too many results that were not relevant to the study; hence, this approach was not used. The first keyword was chosen to direct our search procedure towards farm-level studies, while the second keyword aimed at identifying studies that accounted for the role of education, which could be termed in various ways. The third keyword represented either an environmental dimension targeted for improvement through the adoption of measures or a common technical term for environmental performance as used in the literature. Grouping a technical term as a fourth keyword yielded near zero results; therefore, they were grouped with the various environmental dimension terms. The main aim of this exercise was to initiate a process that would provide us with a set of relevant studies, leading us to further relevant literature through cited work. Accordingly, we used Google Scholar to identify cited papers as well as other articles with similar keyword combinations. The aim was to gather a substantial and diverse body of literature to explore the role of education in farm-level performance.

2.2. Eligibility and Exclusion Criteria

To refine the focus of this review and ensure its relevance to the research objectives while remaining attentive to innovative studies conducted in various regions, we established certain exclusion criteria. The first limitation imposed on the search criteria was (1) the exclusion of papers published before 2000 because of the relative scarcity of research work on environmental performance during this period. Secondly, (2) we excluded studies that were not peer-reviewed scientific articles, such as research articles, review articles, and conference papers. A third (3) exclusion criterion was studies undertaken in countries with primarily subsistence agriculture, typically more concerned about productivity and food security. Fourthly, (4) we excluded studies that analysed data beyond the farm level, such as the regional or national level. We also excluded studies that (5) considered the role of agricultural extension services but not of formal education. Although the search procedure yielded meta-analyses and systematic review studies, (6) we excluded them from our review. Finally, (7) studies exploring solely attitudes rather than intentions to adopt an environmental practice were excluded as well.

Two levels of screening (1: title-abstract-keyword and then 2: full-text screening) were conducted. During the first screening, the abstracts of all articles yielded by the search procedure in Web of Science were read. At this stage, the third exclusion criterion was applied, and the remaining studies were downloaded using EndNote software. The second screening procedure involved the reading of the full text, and the simultaneous application of exclusion criteria (4)–(7). Relevant cited articles and additional articles identified on Google Scholar were subjected to the eligibility and exclusion criteria (4)–(7) as well. Notes were taken to populate Excel tables with information columns to summarise key information. An excerpt of this table is provided in the Supplementary Materials.

3. Results

3.1. Screening and Data Extraction

Figure 1 presents a schematic diagram detailing the selection process for the papers included in the study, highlighting the steps taken to identify eligible articles during the search procedure. The first stage of the search strategy yielded 1264 articles, among which 680 were duplicates, leaving a total of 584 original articles for the first screening procedure.

Reading the abstracts of the remaining papers enabled us to identify 127 relevant articles for full-text reading. Following the application of exclusion criteria (4)–(7), we identified 61 eligible articles. Relevant work cited within these 61 articles, and other papers identified through further exploration on Google Scholar, produced an additional sample of 64 articles, bringing the total number of reviewed articles to 125.

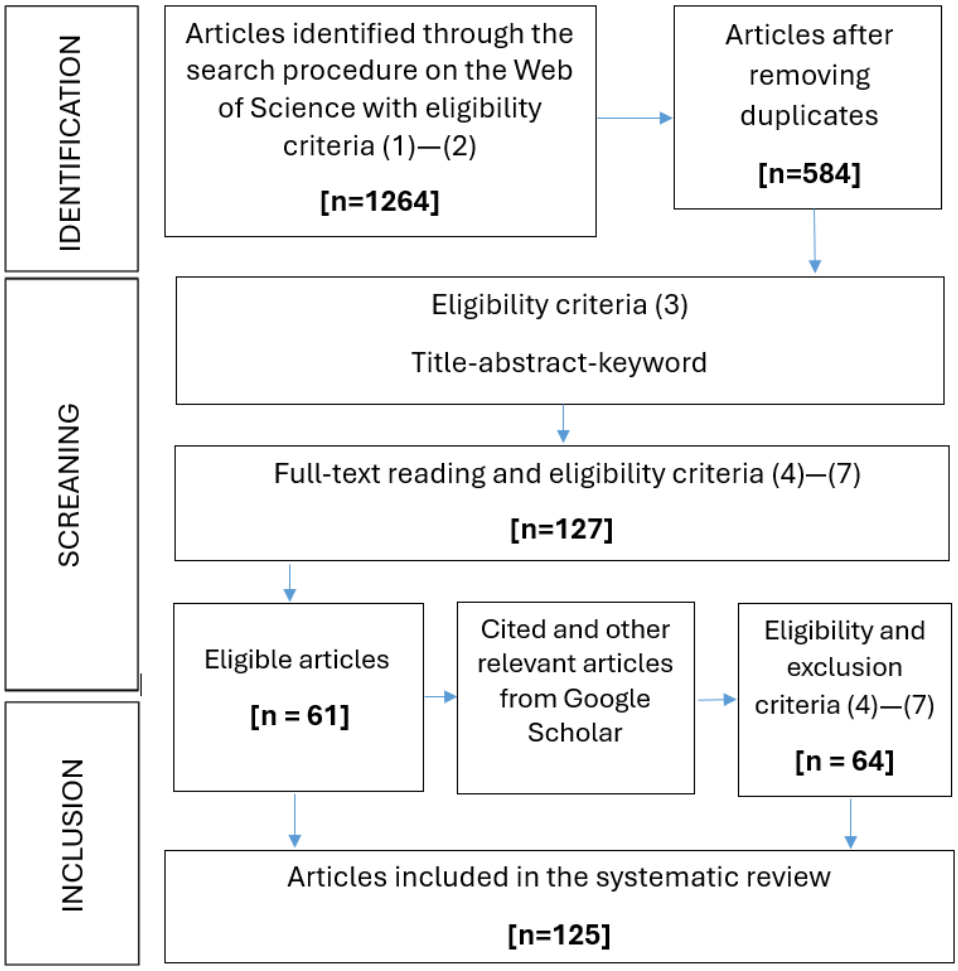

Figure 1. Flow chart for identification, screening, and article inclusion.

We identified 127 studies in the 125 articles; 2 articles explored both actual and intended adoption. Figure 2 illustrates the geographic distribution of the 127 studies. A total of 37, or 29%, pertain to case studies conducted in the EU (excluding Ireland), with Spain leading at 8, while another 37 studies were conducted in China. One of the criteria we followed was the exclusion of developing countries whose policies prioritised food security over sustainability. Although China might still be considered a developing country, its special economic status, paired with its ambitions to become a sustainability leader, has recently prompted a massive transformation of its agricultural sector, which we deemed worthy of examination. A total of 11 studies, or 9%, were conducted in the UK and Ireland, and 18 studies, or 14%, in the United States and Canada. There are 7 studies, or 6%, from New Zealand and Australia, 5 articles, or 4%, from Chile and Brazil, and 11 additional articles, or 9%, from non-EU European countries and Turkey, along with a study that included samples from both EU countries and the UK. The upward trend in the number of studies per year of publication, shown in Figure 3, is a testament to the progression in the interest in environmental performance studies, but also the increasing exploration of the role of sociodemographic factors.

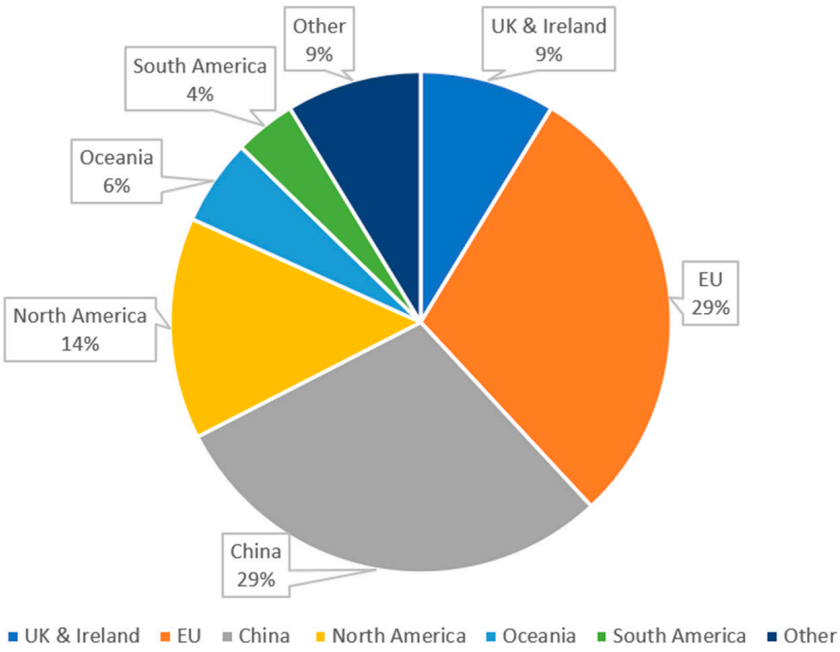

Figure 2. Distribution of reviewed studies included in the analysis, by region.

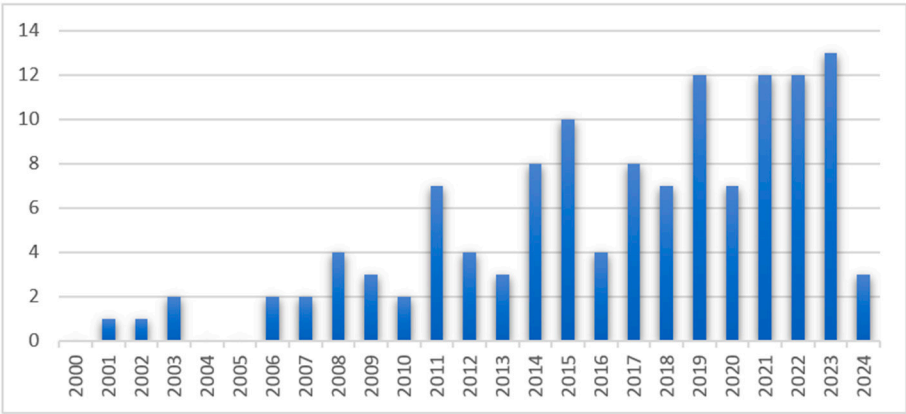

Figure 3. Distribution of reviewed studies included in the analysis, by year of publication.

3.2. Measurement of Formal Education

In 55 out of 127 studies (43%), educational attainment was measured in terms of 'level of education', allocating farmers into categories, for example, by whether they completed primary, secondary, or tertiary formal education. In 28 articles (22%), educational attainment was measured in terms of 'years in education' or 'schooling years', including one that measured formal education as 'age leaving education'. In 32 articles (26%), a single comparison was made in terms of 'low vs. high education', for instance by assessing environmental performance between farmers who had university education and farmers who did not. A total of 18 studies (14%) included 'agricultural education'—11 exclusively and 7 along with formal education. Studies that measured both agricultural education and general formal education did so by using the same education indicator, except for Picazo-Tadeo et al. [19] and Weltin and Hüttel [20], who measured formal education in terms of the 'level of education' and separated farmers into those with and without agricultural education. The variables employed in the analysis were continuous when expressing years in education, ordinal or binary when reflecting increasing levels of education, and binary when a specific level of education was compared to the rest.

3.3. Environmental Behaviour Indicators

We identified three broad types of environmental behaviours investigated: (1) farm-level ecological indicators (EIs), (2) use/adoption of environmental practices or participation in environmental programmes (UAP), and (3) intention to use and adopt environmental practices or participate in programmes (I-UAP). Hence, the first two types of studies (EI, UAP) concern practices currently or recently undertaken by landowners/farmers, while the third (I-UAP) focuses on intended behaviour. In total, there were 31 EI, 61 UAP, and 32 I-UAP studies.

3.3.1. Ecological Indicators

EIs are typically expressed in ratios of input and output quantities of interest [21–24]. Efficiency scores are the most common ecological indicator in this review. The measure links various economic and environmental inputs and outputs and applies them to frontier techniques [19,25,26]. The econometric modelling techniques used to assess the effect of formal education on ecological indicator scores are summarised in the Supplementary Materials.

Efficiency Indicators

Data envelopment analysis is the most employed efficiency modelling method among the reviewed articles [19,27–40], which incorporates several environmental efficiency ratios of interest into a single eco-efficiency or environmental efficiency indicator. These ratios are compared to a benchmark ratio corresponding to the best-performing farm in the sample, which achieves the highest score of one [22]. Other frontier production methods identified in the reviewed studies are the directional distance function method used by Weltin and Hüttel [20] to estimate an eco-efficiency score based on farm deviation from the production possibility frontier, and the stochastic frontier analysis used by Bai et al. [41] and Zhou et al. [42] to assess fertilizer use and irrigation water efficiency, respectively.

Other Indicators

Some researchers used study-specific or other established and previously applied indicators to quantify environmental performance. Backhans, et al. [43] assigned internal and external biosecurity scores for pig farms in Sweden based on practices undertaken. Van Passel et al. [44] formulated a sustainable efficiency index using five forms of capital, among them energy use and nitrogen surplus. Benitez-Altuna et al. [45] devised an ecological intensification practices index. Si et al. [46] used the marginal productivity of using livestock antibiotics. Nave et al. [47] used a pesticide treatment frequency index to classify farmers into intensity categories. Similarly, Backhans, et al. [43] classified farmers into four types based on productivity and nutrient efficiency. Wang et al. [48] used on-farm energy use and greenhouse gas emissions, estimated with life-cycle analysis. Backhans, et al. [43] used fertilizer and chemical pesticide applications as indicators of environmental performance. Zhang et al. [49] used nitrogen application rate, and Ma et al. [50] used the rate of fertilizer use. Shaffer-Morrison and Wilson [51] created the Nutrient Reduction Index based on information on tillage practices, land in cover crops, and years of having small grains in rotation. Other studies [45,52,53] devised adoption indices to assess farms' adoption of conservation practices.

3.3.2. Adoption and Intention to Adopt Environmental Measures

This section focuses on the literature examining the determinants of the intention to adopt and/or the adoption of environmental measures. The largest body of this literature employed regression modelling to identify the socioeconomic determining factors of the adoption and intention to adopt environmental measures. Other studies employed parametric

and nonparametric tests to test for an association between education and environmental performance. The statistical approaches used are summarised in the Supplementary Materials.

3.4. Effect of Education on Farm Environmental Performance

To examine the impact of formal education on environmental performance, we focused on five thematic areas that emerged as key in our review. Accordingly, we grouped environmental performance studies into the following five dimensions: (1) soil and water management, (2) clean energy, (3) biodiversity, (4) organic farming, and (5) agri-environmental programmes. The impact of education across the various dimensions and indicators is highlighted in the following subsections.

3.4.1. Soil and Water Management

In the UK, the eco-efficiency of general cropping farms was better if farm managers had a higher education qualification [34]. In Spain, educational attainment was found to have a positive effect on the eco-efficiency scores of olive farms in the south [28], rain-fed crop systems in the north [19,54], and horticultural family farming in the south-east of the country [31] but not on the environmental efficiency of crop farms in Catalonia [29]. Formal education did not play a role in explaining the eco-efficiency score of Italian crop farms [30], the soil conservation eco-efficiency score in Austria [33], and a public-good provision eco-efficiency indicator of small-scale farms in Poland [32]. Eco-efficiency was higher for farmers with agricultural education across three types of Irish farms: dairy with good land use potential, dairy with average land use potential, and tillage with good land use potential [36]. The level of education of Dutch dairy specialist farmers was the main determinant of improvements in nutrient use efficiency, following the introduction of a nitrogen and phosphate bookkeeping system [35]. Pesticide application intensity was not determined by the level of education of French wheat farmers [47].

Agricultural education was linked to the adoption of non-inversion tillage practices in Ireland [55,56], but not to the adoption of pesticide-free wheat production in Switzerland [57]. Mixed results were reported for England, with Townsend et al. [58] finding that formal education had no association with the choice and extent of reduced tillage, while Wilson [59] reported that college and university graduates were more likely to use 6-metre buffer strips, ponds, wetlands, and minimum tillage. In Sweden, the level of education determined whether farmers used conservation tillage practices [60]. Moreover, farmers who had a university degree were more likely to have already adopted intercropping techniques rather than contemplating adopting them and more likely to have been contemplating adopting them rather than not having contemplated adoption at all [61]. Cover cropping, but not debris management, was more likely to be adopted among olive grove managers in Spain who had agricultural training [62].

The level of education of arable farmers was positively related to the adoption of erosion control measures in silt-loam and sandy-loam regions in Belgium [63]. Among arable farmers who were current users of precision technologies, an agricultural degree affected the intention to adopt new technologies across five European countries [64]. Years in education did not affect the decision to renew participation in climate-smart agricultural practices schemes to improve the soil and water environment in Italy [65]. In a Spanish-based study, the adoption of rainwater harvesting systems for greenhouse irrigation was associated positively with farmers' years in education [66]. Likewise, water conservation measures to reduce soil erosion and nutrient runoff were more likely to be adopted by Finnish farmers with an education level higher than middle school [67].

In the United States, holding a college or graduate degree was linked to a higher nutrient reduction index score [51], as well as building soil organic matter and using

pressure irrigation to conserve water, but not to planting drought-tolerant crops [68]. An agricultural degree increased the uptake of some conservation practices by female farmers [69], while formal education was linked to the uptake of some types of conservation tillage [70,71] but not the choice of cover cropping [72,73] or the adoption of strip-till and reduced tillage [71]. The use of precision agricultural technologies was not linked to having a college education [74]. Whether corn farmers attended college did not affect the adoption of plant tissue testing, or N transformation inhibitors technology [70,75]. Adusumilli and Wang [76] found that college attendance was associated with the adoption of best management practices for water conservation but not for soil conservation.

In China, nutrient use efficiency was poorer among rice farmers with a lower education level [23]. Nitrogen use rates were positively affected by the educational attainment of crop farmers either directly [49] or indirectly [50] but negatively associated among apple farmers unless they had achieved a minimum level of education, beyond which the association became positive [41]. Years in education improved farmers' behaviours and practices around livestock antibiotic use [46]. Further, formal education increased irrigation efficiency in the wheat [27] and maize [42] sectors, while the level of education of crop farmers was positively associated with more efficient application of nitrogen and phosphorous [24]. Educational attainment was a determining factor in adopting conservation tillage techniques [77,78], such as no-tillage farming [79], and straw return [80]. More years in education were associated with a shorter duration between becoming aware of the benefits of conservation tillage and the adoption of minimum tillage, applying weed and pest control, and straw return [81]. Whether or not farmers adopted pasture crop rotation to address soil erosion [82] and measures to prevent desertification [52] in semi-arid areas was not influenced by years in education and the level of education, respectively. Soil testing and use of soil analysis information were more likely to be conducted by farmers who had higher educational attainment in China [83–85] and elsewhere globally [59,69,76,86], with one exception [75]. Two studies identified a positive link between formal education and the use of green and organic fertilizers in China [87,88], while another found no effect [89]. Green control of crop pests and diseases was more likely to be adopted by Chinese farmers with a higher level of education [90].

Having a university degree influenced the probability of adopting several management practices, such as soil moisture monitoring, soil temperature monitoring, rainfall monitoring, and soil fertility tests [91], and soil erosion measures such as tree planting and land retirement, but not water management, soil fixation, and sediment management for livestock farmers in New Zealand [92]. Another study in New Zealand showed that the level of farmers' education predicted whether they implemented a soil management plan and used novel technologies and whether they intended to implement a nutrient management plan [93]. A tertiary qualification was not a determining factor in no-till adoption in Australia [94]. In Chile, formal education did not determine whether vegetable farmers adopted practices to address the negative consequences of agricultural intensification at any stage of production [45], while an agricultural degree was not related to soil conservation behaviour [53].

A higher education degree did not influence the intention to adopt manure separation technology in the Netherlands [95], integrated pest management practices in Serbia [96], and mixed cropping in Germany [97]. On the other hand, the level of education was positively linked to the intention to adopt some permanent grassland-related high-tech innovations, such as remote sensing and the use of software to model GHG emissions, among livestock farmers in Spain [98], and water management practices by Canadian horticultural growers [99].

Chinese farmers with a higher level of education had a more positive attitude toward adopting a winter wheat-fallow policy [100], paying for leguminous green manure [101],

and adopting straw-returning technology [102]. Moreover, the level of education was negatively linked to the willingness to accept ecological compensation for the fallow policy, with a lower compensation value requested by more educated farmers [103]. In Turkey, the willingness to pay for sustainable irrigation training was higher among university graduates [104]; however, farmers with secondary school education were the most willing to pay to apply sustainable land usage practices to avoid salinisation [105].

3.4.2. Clean Energy

Formal agricultural education was linked to adopting and advancing the implementation of climate mitigation measures in the Dutch dairy sector [106]. In Belgium, dairy farms with the highest sustainability scores were more likely to be managed by farmers who had a higher education degree [44]. Potential adopters of on-farm anaerobic digestion in England [107] and Ireland [108] had attained higher levels of formal education. Adopters of renewable energy production measures, such as solar energy, miscanthus, and short-rotation coppice, had a higher level of education than non-adopter English farmers [109]. The installation of solar photovoltaic systems among Dutch arable and livestock farmers was not determined by whether they had a higher education degree [110]. In a German-based study, farmers with a university education were less willing to choose a contract to undertake in-house slurry acidification techniques to reduce livestock ammonia emissions [111].

The carbon emission eco-efficiency of greenhouse tomato growers in Turkey [38] and Swiss dairy farmers [39] was positively influenced by years in education and agricultural qualifications, respectively, but this was not the case for Greek dairy and sheep farmer [40] and Scottish dairy [37] farmers. In the United States, a higher level of education affected the share of cropland that farmers were willing to convert to perennial switchgrass for energy production [112], but not the willingness to grow annual energy crops [113]. The adoption of cleaner capital-intensive production measures was more likely among Chilean dairy farmers with higher levels of education [114].

The level of education in the Chinese rice sector was positively associated with the probability of adopting a lower-carbon technology and with the number of technologies adopted [115]. In other Chinese studies, formal education was positively associated with the willingness to install biogas digestors in the beef sector [116], the utilisation of biogas made from crop straw [117], and the willingness to reuse agricultural waste for carbon emission abatement [118]. Years in education positively affected the probability of recycling pesticide packaging among Chinese apple growers [119] and the adoption of on-farm solar energy production technology [120]. On the other hand, years in education did not affect the energy consumption and greenhouse gas emissions efficiency of rice–wheat rotation system farms [48].

3.4.3. Biodiversity

Adopting sustainable intensification measures to increase biodiversity in peaty areas was more likely among German farmers with an agricultural degree [20]. The higher the agricultural qualification of Swiss farmers, the higher the share of land they registered as Ecological Focus Areas for landscape interventions to benefit biodiversity [121]. In a France-based study, the intention to participate in agri-environmental biodiversity offset schemes was higher among those farmers who had completed a longer secondary education [122]. Although Belgian farmers with a high school certificate were more willing to introduce permanent features in their fields, such as trees and hedgerows, they did not differ in their interest in expanding their field margins compared to lower-educated farmers [123]. In Sweden, research indicated that pig herd managers with a higher education degree achieved a better internal biosecurity score [43].

In Canada, more years in education were linked to a higher probability of engaging in a conservation programme on non-working land to protect natural heritage features [124]. In the United States, cattle farmers with at least some college education were more willing to engage in afforestation to reduce carbon emissions [125]. In Australia, landholders who engaged in native vegetation planting had a higher level of education [126]. However, the level of education was not associated with participation in biodiversity conservation programmes [127].

In China, years in education positively determined the likelihood of participating in an international large-scale forest project, unlike participating in a national project [128]. The intention to adopt best management practices in slopy cropland, such as no-till tillage, conversion of farmland to forest or grassland, and the establishment of grassed waterways or field borders, was positively associated with the level of education [129]. Although willingness to participate in a cropland-to-forest conversion programme was higher among farmers with high school education [130], a more recent study found that years in education negatively affected the propensity to participate in the 'grain to green' programme, which entailed the conversion of sloping hills to forests [131].

3.4.4. Organic Agriculture

Owners of organic olive groves in Greece [132] had completed more years in education, as did German organic cereal farmers (ibid.). The probability of producing organically rather than conventionally in Norway increased if the farmer had a university degree or an agricultural education [133]. In Greece, farmers' level of education determined whether they partially or fully adopted organic crop farming [134], but holding a tertiary degree did not influence their willingness to participate in an organic farming subsidy scheme in the vinicultural sector [135]. Canadian organic grain farmers who attended university were not more likely to be partial or complete adopters, nor did they have a larger share of land under organic production [136]. Having a higher education degree was not a determining factor of whether Irish drystock farmers were adopters of organic farming, nor their stage of adoption [137]. In Turkey, educational attainment beyond primary school was found to be a negative determinant of willingness to produce organic milk [138].

3.4.5. Agri-Environmental Programmes

The adoption of a nitrate reduction programme was positively affected by whether farmers had qualifications beyond a primary school certificate in Greece [139]. In Slovenia, farmers who had finished secondary school were more likely to participate in the Agricultural Environmental Climate Scheme than farmers of both a lower and a higher education level [140]. A higher level of education positively determined participation in agri-environmental programmes in Poland [141] but characterised resistant non-participants in Italy [142]. A study across five European regions found a positive association between having completed an agricultural degree and participation in agri-environmental schemes [143]. The level of education was not linked to participating in Nova Scotia's Environmental Farm Plan targeting intensive farming in Canada [144,145]. Similarly, years in education did not determine participation in ecosystem services payment programmes in Brazil [146,147].

3.5. Vote Count

Vote-count analysis provides a more transparent account of the links between education and environmental performance, since literature reviews might tend to focus their lenses on significant results, both negative and positive. We explored the statistical significance of the effects and associations reported in the literature by conducting vote counting of positive, negative, and insignificant relationships. In the case of regression analysis, we extracted the sign and coefficient values corresponding to the effect of education variables

on an environmental performance indicator. In the case of studies conducting parametric and nonparametric tests, we extracted statistically significant associations when there was a clear indication about the association's direction. We considered associations statistically significant if $p \leq 0.05$, since there were studies that did not report results at a higher level. Four studies were excluded from the vote-count procedure; one of them solely reported descriptive statistics, while three others did not provide any indication of the direction of the association. Therefore, the total number of studies considered in the vote count is 123. The number of unique education variables employed is 175, while the total effects and associations tested are 405. More information can be found in the excel document of the Supplementary Materials.

Table 1 reports the results of this exercise at the study level, in an attempt to quantify the overall conclusions drawn from the literature review, while Table 2 does so at the variable level. In both tables, we report percentages across the various thematic dimensions explored in the literature review, and additionally at the environmental behaviour and education indicator levels. Chi-squared tests showed significant differences in the proportions of studies with only or mainly positive, at least one positive, and no positive effects and associations across thematic dimensions ($\chi^2 = 80.122$; $p < 0.01$), education indicators ($\chi^2 = 248.800$; $p < 0.01$), but not environmental performance indicators ($\chi^2 = 5.32$; $p = 0.256$). Similarly, there were significant differences in the proportions of negative, insignificant, and positive effects and associations across thematic dimensions ($\chi^2 = 37.602$; $p < 0.01$) and education indicators ($\chi^2 = 86.853$; $p < 0.01$) but not environmental performance indicators ($\chi^2 = 4.440$; $p = 0.350$).

Table 1. Reporting significance results at the study level.

Reported Association of Education with Environmental Performance	# Studies	95% Statistical Significance		
		No Positive Effects and Associations	At Least One Positive Effect or Association	Only or Mostly Positive Effects and Associations
Per thematic dimension				
Soil and water management (Europe)	**31**	**14** (45%)	**11** (35%)	**6** (19%)
Soil and water management (Other)	**42**	**14** (33%)	**8** (19%)	**20** (48%)
Clean energy	**20**	**8** (40%)	**2** (10%)	**10** (50%)
Biodiversity	**13**	**3** (23%)	**5** (38%)	**5** (38%)
Organic agriculture	**8**	**5** (63%)	**2** (25%)	**1** (13%)
Agri-environmental programmes	**9**	**5** (56%)	**1** (11%)	**3** (33%)
Per environmental behaviour indicator				
Ecological indicator (EI)	**30**	**12** (40%)	**8** (27%)	**10** (33%)
Use, adoption, participation (UAP)	**61**	**23** (38%)	**12** (20%)	**26** (43%)
Intention to use, adopt, participate (I-UAP)	**32**	**14** (44%)	**9** (28%)	**9** (28%)
Per education indicator (The total number of education indicators is larger by 7 due to double counting of studies that measured both agricultural and general formal education)				
Agricultural education (any type)	**17**	**5** (29%)	**7** (41%)	**5** (29%)
Level of education (ordinal/categorical)	**36**	**14** (39%)	**2** (6%)	**20** (56%)
Level of education (multiple dummies)	**17**	**3** (18%)	**12** (71%)	**2** (12%)
Low vs. high education level (one dummy)	**32**	**25** (78%)	**2** (6%)	**5** (16%)
Years in education (continuous)	**28**	**8** (29%)	**2** (7%)	**18** (64%)
Total (Total for each column only applies to thematic dimension and environmental behaviour indicators. This is due to double counting of education indicators and differences in the overall study conclusions arising when separating the effects and associations relevant to agricultural education from those relevant to general formal education)	**123**	**49** (40%)	**29** (24%)	**45** (37%)

As seen in Table 1, European studies falling into the soil and water management dimension, along with organic agriculture, were less likely to produce solely positive results, although the latter group, along with studies falling in the agri-environmental programmes dimension, were the only ones with over 50% of studies producing no positive

results. The percentage of studies observing at least one positive effect or association is strikingly larger for studies using a 'level of education' indicator, which involves employing a binary variable for each category of education. The percentages across studies at the environmental behaviour indicator level are fairly similar. Table 2 presents a more balanced image, also backed by the lower chi-squared values. Across the two tables, the most notable result is the mere 9% positive effects and associations observed by studies that employed a single binary education variable. At the study level, the share is 13%.

Table 2. Reporting significance results at the variable level.

Reported Association of Education with Environmental Performance	# Education Variables	# Effects or Associations Tested	95% Confidence		
			Neg. (−)	Insig.	Pos. (+)
Per thematic dimension					
Soil and water management (Europe)	53	**97**	**2** (2%)	**74** (76%)	**21** (22%)
Soil and water management (Other)	58	**165**	**2** (1%)	**95** (58%)	**68** (41%)
Clean energy	23	**59**	**2** (3%)	**28** (47%)	**29** (49%)
Biodiversity	20	**50**	**0** (0%)	**28** (56%)	**22** (44%)
Organic agriculture	9	**16**	**1** (6%)	**11** (69%)	**4** (25%)
Agri-environmental programmes	12	**18**	**1** (6%)	**12** (67%)	**5** (28%)
Per environmental behaviour indicator					
Ecological indicator (EI)	46	**96**	**2** (2%)	**67** (70%)	**28** (29%)
Use, adopt, participate (UAP)	82	**216**	**3** (1%)	**124** (57%)	**88** (41%)
Intention to use, adopt, participate (I-UAP)	47	**93**	**3** (3%)	**57** (61%)	**33** (35%)
Per education indicator					
Agricultural education (any type)	26	**78**	**2** (3%)	**46** (59%)	**30** (38%)
Level of education (ordinal/categorical)	36	**105**	**0** (0%)	**43** (41%)	**62** (59%)
Level of education (multiple dummies)	51	**110**	**1** (1%)	**84** (76%)	**25** (23%)
Low vs. high education level (one dummy)	32	**67**	**3** (4%)	**58** (87%)	**6** (9%)
Years in education (continuous)	30	**45**	**2** (4%)	**17** (38%)	**26** (58%)
Total	**175**	**405**	**8** (2%)	**248** (61%)	**149** (37%)

4. Discussion

The aim of this paper was to synthesise the findings of applied work, which has explored, among others, the role of formal education in environmental performance. We show that acquiring or extending formal education can have a positive impact on farmers' environmental performance, as shown previously on the global [16,17,148–150] and European stages [151–153]. In this present review, 60% of all studies reported at least one positive effect or association between farmers' education and environmental performance. However, in line with the consensus in the literature, the evidence is far from conclusive. This becomes more obvious when moving away from the evidence drawn from the literature review. Vote-count analysis at the variable level, as opposed to vote-count analysis at the study level, which closely reflects the conclusions of the literature review, shows a different picture. Accordingly, only 37% of all tested effects and associations were positive and statistically significant. This is of a similar magnitude to meta-analysis studies on conservation practices [148,149] and participation in agri-environmental schemes [151].

We posit that the way educational attainment is measured and represented in the analysis can affect the conclusions of a study, an observation also made by Burton [154], but also the conclusions of meta-analysis. Take, for example, the case of studies exclusively exploring the effect of attaining tertiary qualifications in comparison with other educational achievements. This approach will likely conceal the distinct impacts that other educational levels have on environmental performance. Indeed, Tables 1 and 2 reveal that a single binary representation of education was almost always associated with an insignificant result. In the literature review, we identified several cases where education exhibited fluctuating effects. For instance, [148,149], who used a continuous quadratic 'years in education' variable, demonstrated an initially positive impact on the probability of adopting organic production in Greece and Germany, followed by an opposite effect beyond a country-specific threshold

point. An inverse U-effect of 'years in education' was observed by [148,149] on fertiliser use efficiency in China, as well as by [148,149] who used multiple 'level of education' binary variables to assess Spanish farms' eco-efficiency. In contrast, [148,149] showed that Chinese farmers with no education were less willing to participate in afforestation, while farmers with some education and farmers with a college degree did not differ in their preference, showing strictly positive but non-linear effects. The latter result likely captured the high marginal benefit of educational attainment during its early stages. This is reflected in our analysis, which showed that the use of the 'years in education' indicator, mainly by studies conducted in China, was associated positively with environmental performance 64% of the time. However, the use of a continuous variable inevitably assumes constant returns from education.

It is likely that single binary variables were chosen precisely because a more detailed representation of educational attainment did not generate significant results. This might be supported by the fact that among studies employing multiple binary variables for levels of education, nearly all reported at least one positive result, as seen in Table 1. Input from studies accounting for multiple levels of education is desirable as it provides insight into the benefits that education can have for different segments of farmers. However, this has the unintended consequence that vote-count analysis at the variable level records a higher proportion of insignificant effects and associations than would be recorded if it recorded one insignificant result for a single dummy variable. Hence, researchers must be careful with the interpretation of vote-count analysis results, as the results will depend to some extent on the type of education variables employed in studies.

Another observation from the literature review is that researchers often based their positive conclusions about the role of education on a lower-than-5% statistical significance level. For instance, the literature review conclusions drawn from some studies [60,61,67,76,120,123,128,133,142] contrast with the reported insignificant effects and associations at the 5% level, the significance level typically used in meta-analysis studies. In fact, there were cases where authors drew conclusions, albeit on the cautious side, based on the sign rather than significance, which can be justified on some grounds, particularly when an opposite result occurs. Therefore, this literature review draws a larger share of studies inferring a positive role for education than the numbers in Table 1 would suggest. This discrepancy creates a divergence between the conclusions emerging from the literature review and the inference from vote-count analysis, particularly at the variable level.

Researchers sometimes differentiate between general formal education and specialised agricultural education to shed light on their distinct roles in affecting environmental performance, as noted by [60,61,67,76,120,123,128,133,142]. Accounting solely for agricultural education was associated with at least one positive result [36,39,55,56,62,64,69,106,121], except in one case [57], while among articles that accounted for both general education and agricultural education [19,20,28,53,54,133,143], a varied combination of associations was observed. This is not reflected in the overall result for the 'agricultural education' indicator in Table 1, which illustrates instead a more balanced picture. We highlight the importance of developing distinct hypotheses about the role of the two types of education. An undesirable effect can be the confounding of the two respective variables, as inferred by Rodríguez-Entrena and Arriaza [62], particularly when they are not mutually excluded. Confounding can also be observed with other factors affecting environmental performance to which education is strongly linked, particularly age and experience, offering a challenge to the applied researcher [154].

Accounting for other sources of knowledge and information can reveal a more far-reaching role for education since access to information and information utilisation are fa-

cilitated by skills obtained with educational attainment. This is demonstrated in a Greek study [134], which showed that the decision to convert to organic farming was positively correlated with actively or passively seeking farming-related information among better-educated farmers. Alternative sources of knowledge and information, for instance, consultant recommendations [94], agricultural extension [82,94], training [82], field days, workshops and information sessions [145], and contractor advice [75], were shown to influence environmental performance, unlike formal education, although other studies [52,81] yielded a different mix of results. We highlight the possibility that the utilisation of alternative sources of information might be positively linked to formal education while exerting a more direct effect on environmental performance than the latter. Nonetheless, demonstrated knowledge about the beneficial effects of environmental measures does not guarantee superior environmental behaviour, neither for farmers with a higher general formal education [95] nor for farmers with an agricultural degree [53].

4.1. Policy Implications

The findings from this review offer important implications for policy development. The inconclusive results suggest that policies may need to be specific, targeting groups of farmers, such as those who have no formal agricultural training, those with some knowledge and those with high qualifications, as a one-size-fits-all policy may not be effective. In short, policies should be tailored to match the needs, preferences, knowledge, and perceived abilities characterising farmers of different educational attainment. For instance, although results-based biodiversity schemes were more likely to attract farmers with high levels of agricultural education in Switzerland, action-based schemes were the preferable option for farmers with lower-level agricultural training [121]. It is often the case that specific groups of farmers are more likely to engage in a new measure; for example, voluntary measures to avoid salinisation would more likely materialise if targeting specifically farmers who had a secondary education in Turkey. Likewise, policies might be effective only among farmers of a specific education level. Increasing computer skills and expanding cooperative membership in China would reduce livestock antibiotic use [46] and increase packaging recycling [119], respectively, albeit only among farmers with more than primary education.

Diverse motivations for farming can prompt a different response to incentives, even among farmers of the same educational attainment. In this case, efforts should be directed towards areas where engagement is most likely. For instance, though highly educated active farmers in Finland were the most likely to adopt voluntary conservation measures, equally educated passive landowners had the opposite response [67]. Studies employing structural modelling provide insight into factors such as attitudes and norms that mediate the relationship between education and environmental performance. Ex ante understanding of the diverse motivations for farming [93,142,155] and the social and personal norms [87,90] characterising different education cohorts can potentially predict the acceptance of environmental programmes and measures. For example, although promoting conservation measures by inducing the ascription of responsibility across the educated classes would be successful in China [102], promoting nutrient efficiency measures by appealing to social acceptance has proven unsuccessful for highly educated farmers in The Netherlands [35].

Perceived behavioural control to undertake conservation measures might influence environmental performance. The limited evidence in our review [96,97] does not support this, suggesting other factors are more directly relevant. Nonetheless, regardless of farmers' confidence, policies should account for barriers faced by the less educated, which were shown to affect the uptake of environmental measures in China when they involved international collaboration [128] and labour transfer to other industries [100]. Government

policies established to encourage people to remain in formal learning might need to target specific types of farmers for better returns. In the case of China, for instance, efforts to reduce the overapplication of fertiliser should focus on training farmers who are expected to depend less on farm income and those of moderate educational attainment as they were less inclined to rely on experience [49]. Perceived ecological benefits positively influenced attitudes towards mixed cropping in Germany and, consequently, intention to adopt. Therefore, the role of education can be multifaceted, acting as a vehicle in facilitating the diffusion of scientific knowledge and altering attitudes towards changes from the status quo that might at first appear to be at odds with conventional knowledge.

4.2. Limitations

In interpreting these results, it should be emphasised that the study has limitations. While we adhered to the PRISMA methodology, we acknowledge that our initial reliance on education-related keywords might have resulted in inadvertently missing relevant publications. This could have happened because studies were more likely to mention results about education in their abstract if they accorded with their working hypothesis. For instance, it appears that some of our included studies may have considered education but excluded it from parts of their analysis (e.g., [44,139,146]), possibly to enhance model performance or highlight statistically significant outcomes. Although we cannot fully account for such omissions, we sought to minimise bias from the initial screening by thoroughly reviewing selected articles and identifying relevant citations for deeper exploration. Moreover, as highlighted by our reviewers, our search procedure should have been better linked to the theoretical underpinnings of environmental performance. Since our aim was to emulate a catch-all approach that includes diverse environmental practices, the third set of keywords can appear quite abstract. Our keyword approach was chosen to initiate our search for relevant work, leading us to identify through citations an even larger number of studies. Moreover, although our resulting thematic dimensions were not ex ante determined, they represent the most often investigated types of farm-level interventions. Hence, we do contend that our review draws adequately from the targeted literature. Another limiting factor stems from our quest to draw from a large, heterogeneous literature employing a diverse set of methodological and statistical approaches, applied to distinct socioeconomic and cultural backdrops. To this end, the results displayed in Tables 1 and 2 should be used as a general reference and a potential guide for more complex analyses. For instance, meta-regression analysis accounting for these heterogeneities, including the way education is measured, can provide more convincing evidence about the role of education in environmental performance.

5. Conclusions

This literature review explored the effect of formal education on environmental performance at the farm level. To achieve our objective, we reviewed the extant literature by employing the PRISMA methodology and provided an overview across five thematic dimensions. Studies on conservation measures aimed at improving soil and water quality were the most common thematic dimension, representing well over half the reviewed studies. We additionally categorised studies according to the environmental behaviour indicator used: (1) farm-level ecological indicators, (2) use and adoption of environmental practices, and (3) intention to use and adopt environmental practices. Half of these articles explored the socioeconomic determinants influencing the decision to use and adopt environmental measures.

Overall, 60% of studies reported at least one positive association between formal education and environmental performance, including 38% of studies reporting solely

or mostly positive association(s). A vote-count analysis at the variable level showed a different picture, with only 37% of the reported effects and associations being positive. This difference mirrors, to an extent, the divergence in the conclusions drawn from the literature review compared to the vote-count analysis. We suggest that the lower share of positive effects and associations in the vote-count analysis at the variable level does not negate the important role of education, while a simplistic interpretation risks overlooking the importance of formal education for specific segments of the farmer population.

Overall, the types of formal education considered, the way of constructing the respective indicators, and the consideration of other factors relevant to each case study can affect the conclusions about the role of education, both at the study and the meta-analysis level. Renewed empirical research at the farm level that will consider the role of education more extensively has the potential to offer enhanced insight for future meta-analysis. Moreover, it will provide justification for targeted approaches in promoting the uptake of environmental measures by farmers with different levels of education and in encouraging the advancement of educational attainment.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/su17020796/s1, An excel spreadsheet with relevant information extracted from the literature review and data used in the vote-count analysis.

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